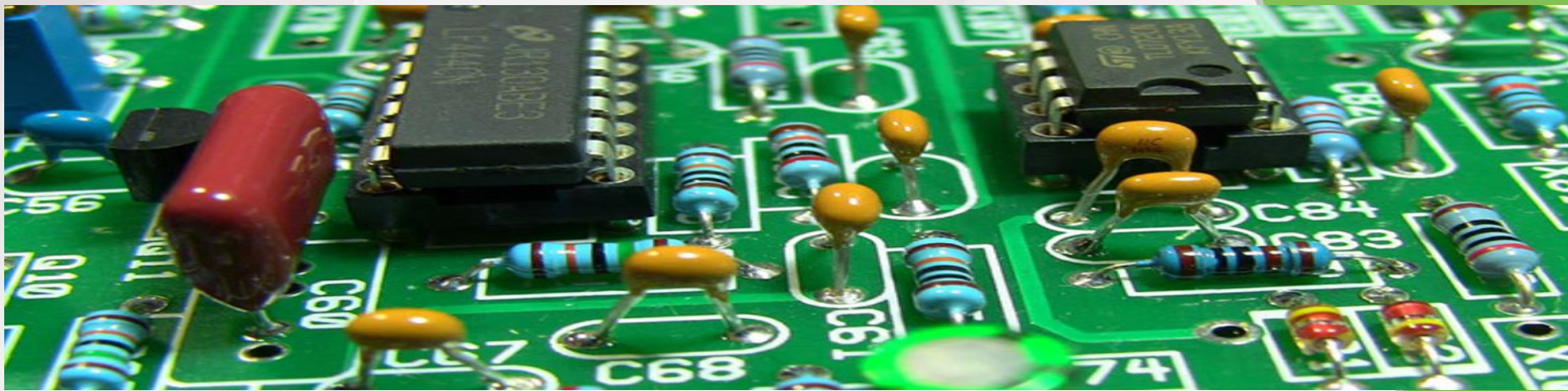




62732 Analog electronics



Technical University of Denmark

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62732 IoT

Slide 1

LT- spice frekvens simulering



Realization of lowpass filter

- The second order transfer function can be written by using the voltage divider rule:

$$T(s) \equiv \frac{V_o}{V_i} = \frac{Z_2}{Z_1 + Z_2} = \frac{Y_1}{Y_1 + Y_2} = \frac{1/sL}{(1/sL) + sC + (1/R)}$$

$$= \frac{1/LC}{s^2 + s(1/CR) + (1/LC)}$$

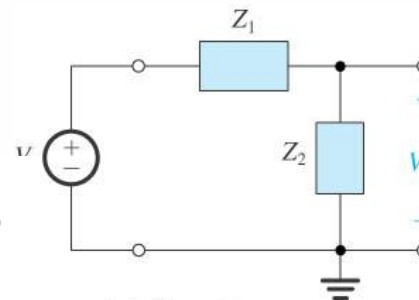
$$T(s) = \frac{a_2 s^2 + a_1 s + a_0}{s^2 + (\omega_0/Q)s + \omega_0^2}$$

$$\omega_0^2 = 1/LC \quad \omega_0/Q = 1/CR$$

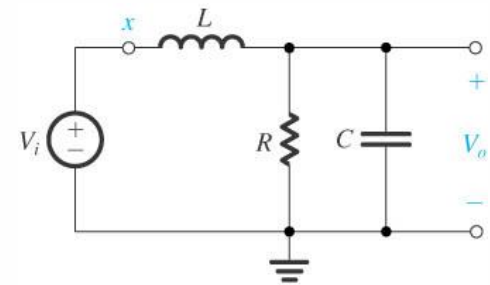
Or:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad Q_p = \frac{R}{2\pi f_0 L} \quad Q_p = 2\pi f_0 CR$$

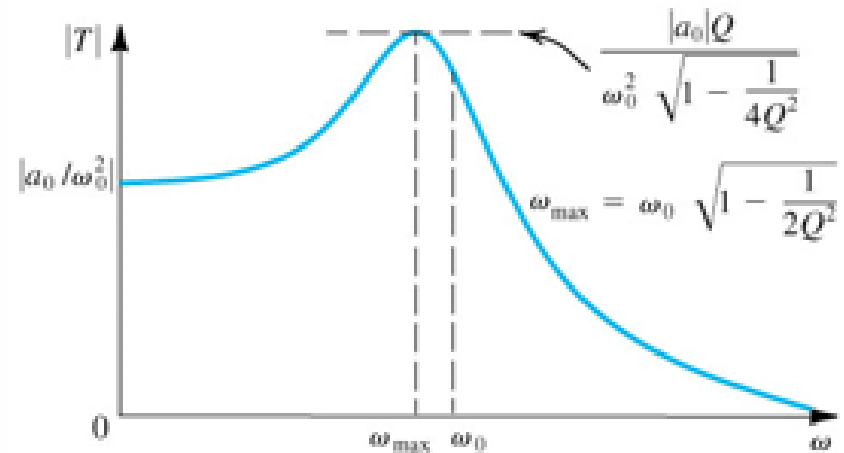
$$T(s) = \frac{V_o(s)}{V_i(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$$



(a) General structure



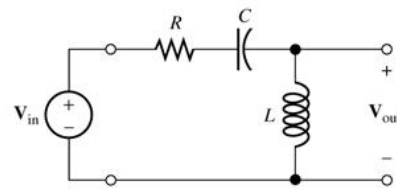
(b) LP



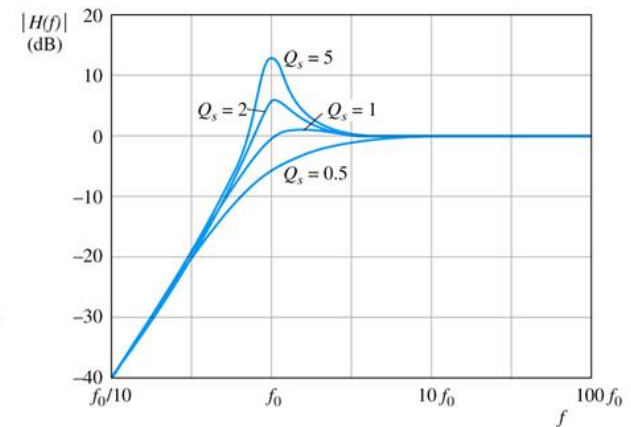
Try: $Q=100$ – how large is the peak compared to the value at $\omega=0$

Realization of high-pass filter

□ Example:

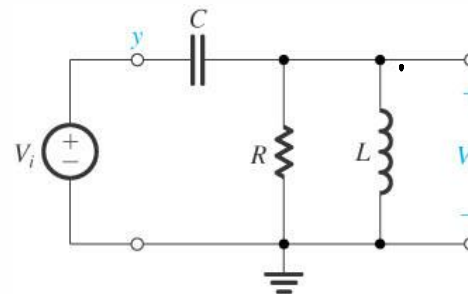


(a) Circuit diagram

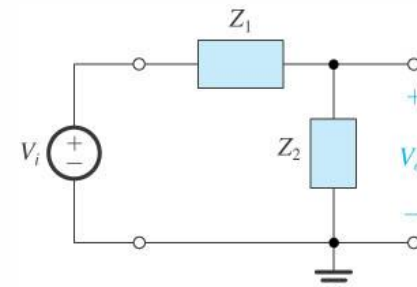


(b) Transfer-function magnitude

□ Another example:



(c) HP



(a) General structure

$$T(S) = \frac{Z_2}{Z_2 + Z_1} = \frac{Y_1}{Y_1 + Y_2} = \frac{Cs}{Cs + \frac{1}{Ls} + \frac{1}{R}} = \frac{RLCs^2}{RLCs^2 + R + Ls} = \frac{s^2}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

$$T(s) \equiv \frac{V_o}{V_i} = \frac{a_2 s^2}{s^2 + s(\omega_0/Q) + \omega_0^2}$$

$$\omega_0^2 = 1/LC$$

$$\omega_0/Q = 1/CR$$

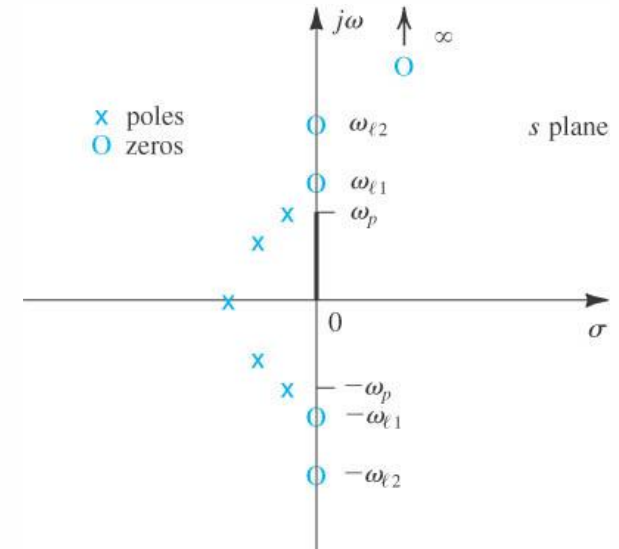
$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$Q_p = \frac{R}{2\pi f_0 L}$$

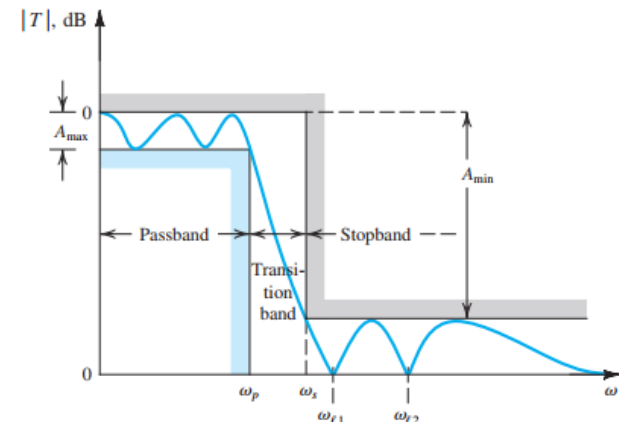
$$Q_p = 2\pi f_0 CR$$

Poles position for a low-pass filter

- In the filter stopband the transmission is required to be zero (small),
 - zeros are usually placed on the $j\omega$ axis at stopband frequencies.
- Example zero transmission (infinite attenuation) at two stopband frequencies ω_1 and ω_2 .
 - zeros at $s = +j\omega_1$ and $s = +j\omega_2$.
 - Complex zeros occur in conjugate pairs, i.e. $s = -j\omega_1$ and $s = -j\omega_2$.
- The numerator polynomial must have the factors:
 $(s+j\omega_1)(s-j\omega_1)(s+j\omega_2)(s-j\omega_2)$ or
 $(s^2 + \omega_{l1}^2)(s^2 + \omega_{l2}^2)$
- $N - M$ zeros at $s = \infty$.
- Stable filter requires all poles must lie in the left half of the s plane ($p_1, p_2, \dots, p^N$ must all have negative real parts).



$$T(s) = \frac{a_4(s^2 + \omega_{l1}^2)(s^2 + \omega_{l2}^2)}{s^5 + b_4s^4 + b_3s^3 + b_2s^2 + b_1s + b_0}$$

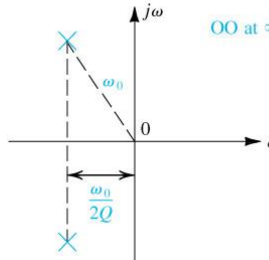
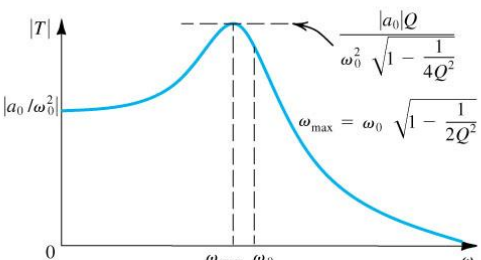
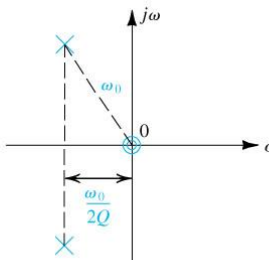
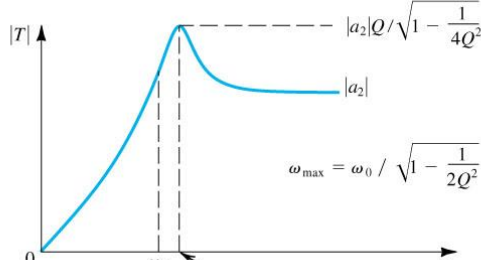
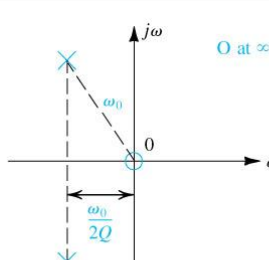
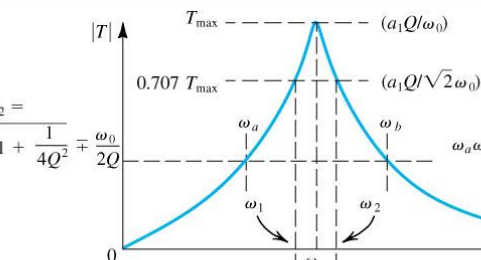


SECOND-ORDER FILTER FUNCTIONS



Second-Order Filter Functions

- Examples:
- Low-pass the two transmission zeros are at $s = \infty$.
- The magnitude response can have a peak
- The peak occurs only for $Q > 1/\sqrt{2}$.

Filter Type and $T(s)$	s-Plane Singularities	$ T $
(a) Low pass (LP) $T(s) = \frac{a_0}{s^2 + s\frac{\omega_0}{Q} + \omega_0^2}$ DC gain = $\frac{a_0}{\omega_0^2}$		
(b) High pass (HP) $T(s) = \frac{a_2 s^2}{s^2 + s\frac{\omega_0}{Q} + \omega_0^2}$ High-frequency gain = a_2		
(c) Bandpass (BP) $T(s) = \frac{a_1 s}{s^2 + s\frac{\omega_0}{Q} + \omega_0^2}$ Center-frequency gain = $\frac{a_1 Q}{\omega_0}$		

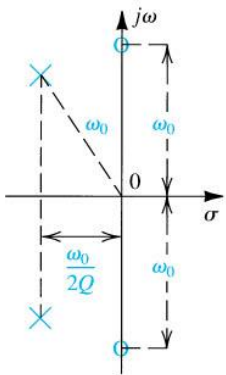
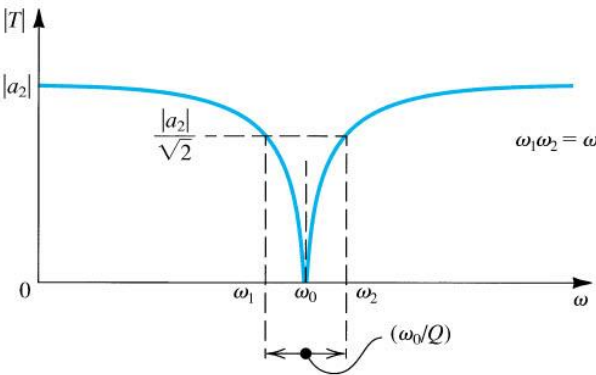
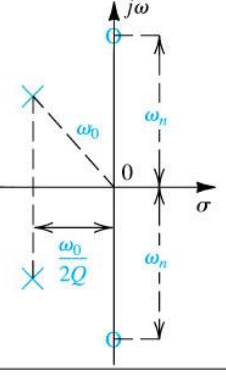
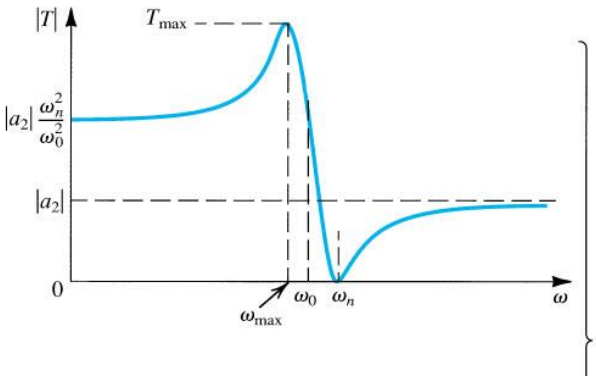
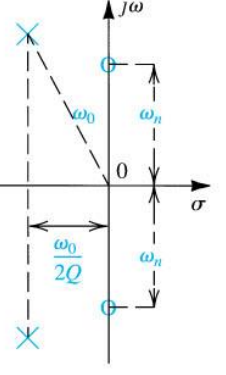
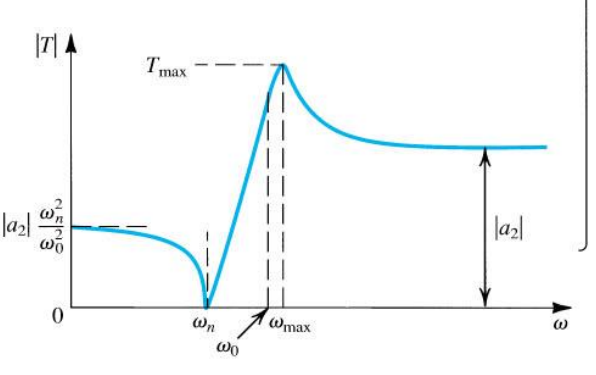
Filter Type and $T(s)$	s-Plane Singularities	$ T $
(d) Notch  $T(s) = a_2 \frac{s^2 + \omega_0^2}{s^2 + s \frac{\omega_0}{Q} + \omega_0^2}$ DC gain = High-frequency gain = a_2		
(e) Low-pass notch (LPN) $T(s) = a_2 \frac{s^2 + \omega_n^2}{s^2 + s \frac{\omega_0}{Q} + \omega_0^2}$ $\omega_n \geq \omega_0$ DC gain = $a_2 \frac{\omega_n^2}{\omega_0^2}$ High-frequency gain = a_2		 $\omega_{\max} = \omega_0 \sqrt{\left(\frac{\omega_n^2}{\omega_0^2}\right)\left(1 - \frac{1}{2Q^2}\right) - 1}$
(f) High-pass notch (HPN) $T(s) = a_2 \frac{s^2 + \omega_n^2}{s^2 + s \frac{\omega_0}{Q} + \omega_0^2}$ $\omega_n \leq \omega_0$ DC gain = $a_2 \frac{\omega_n^2}{\omega_0^2}$ High-frequency gain = a_2		 $T_{\max} = \frac{ a_2 \omega_n^2 - \omega_{\max}^2 }{\sqrt{(\omega_0^2 - \omega_{\max}^2)^2 + \left(\frac{\omega_0}{Q}\right)^2 \omega_{\max}^2}}$

Figure 12.16 (Continued)

(g) All pass (AP)

$$T(s) = a_2 \frac{s^2 - s \frac{\omega_0}{Q} + \omega_0^2}{s^2 + s \frac{\omega_0}{Q} + \omega_0^2}$$

Flat gain = a_2

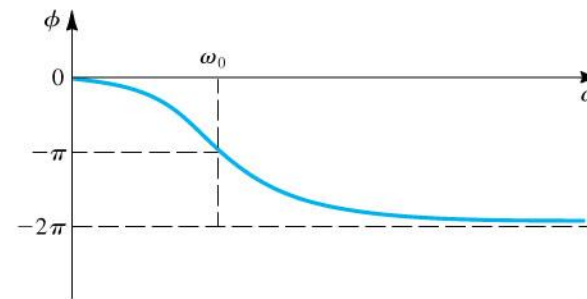
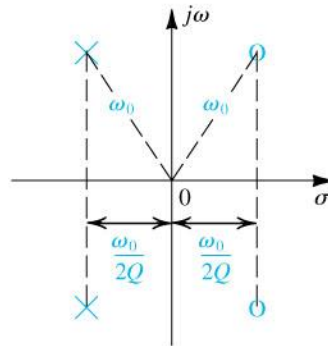
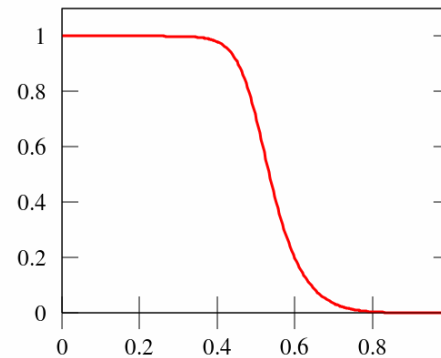


Figure 12.16 (Continued)

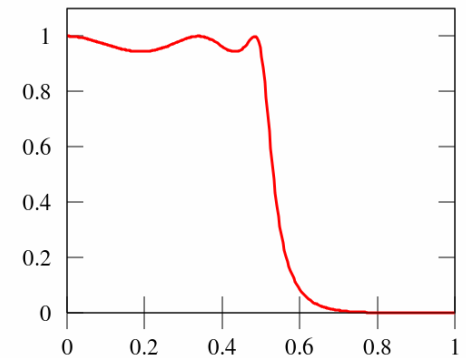
Filter Approximations Overview

- Some classic analogue filter types are Butterworth, Chebyshev, and Elliptic filters.
- There is no ideal filter:
 - Butterworth: Flattest pass-band but a poor roll-off rate.
 - Chebyshev: Some pass-band ripple but a better (steeper) roll-off rate.
 - Elliptic: Some pass- and stop-band ripple but with the steepest roll-off rate.

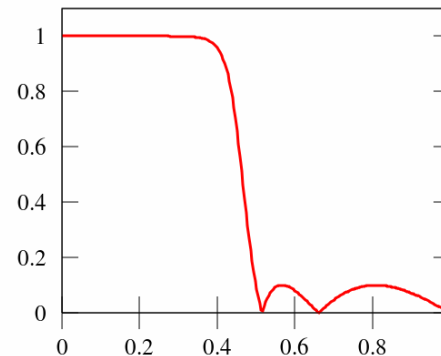
Butterworth



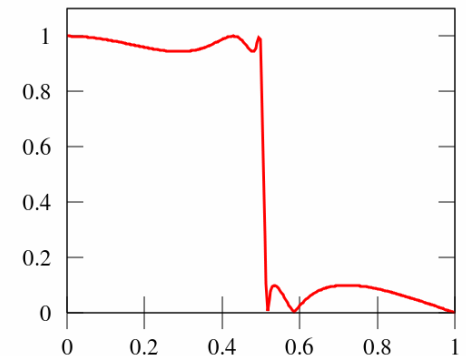
Chebyshev type 1



Chebyshev type 2

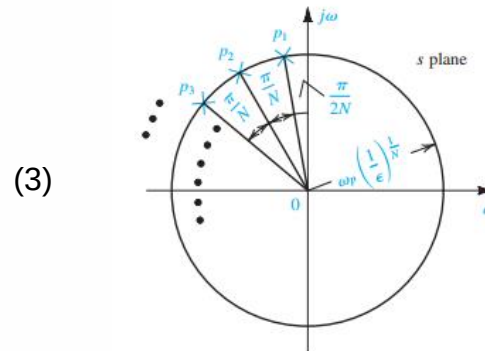


Elliptic



Butterworth filter

- Magnitude response of a Butterworth filter.
- This filter exhibits a monotonically decreasing transmission
- All transmission zeros are at $\omega = \infty$, i.e. it is an all-pole filter.
- The magnitude function for an N th-order Butterworth:
- The attenuation of the Butterworth filter:
- Design procedure
 - Find ϵ from (1)
 - Find N from (2)
 - Determine the poles position from (3)
 - Write the transfer function (4)



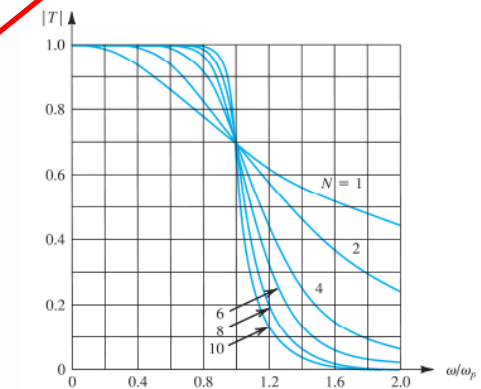
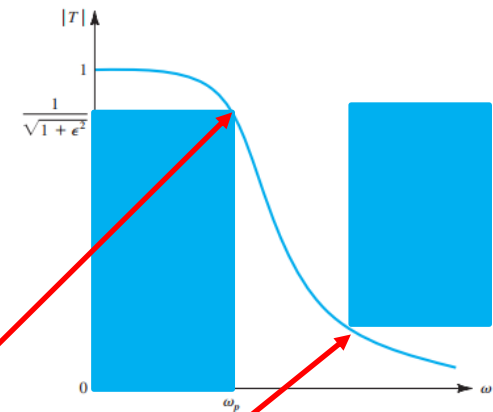
$$|T(j\omega)| = \frac{1}{\sqrt{1 + \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^{2N}}}$$

$$|T(j\omega_p)| = \frac{1}{\sqrt{1 + \epsilon^2}}$$

(1) $A_{\max} = 20 \log \sqrt{1 + \epsilon^2}$

(2) $A(\omega_s) = -20 \log \left[\frac{1}{\sqrt{1 + \epsilon^2 (\omega_s/\omega_p)^{2N}}} \right]$
 $= 10 \log \left[1 + \epsilon^2 (\omega_s/\omega_p)^{2N} \right]$

(4) $T(s) = \frac{K\omega_0^N}{(s-p_1)(s-p_2)\cdots(s-p_N)}$

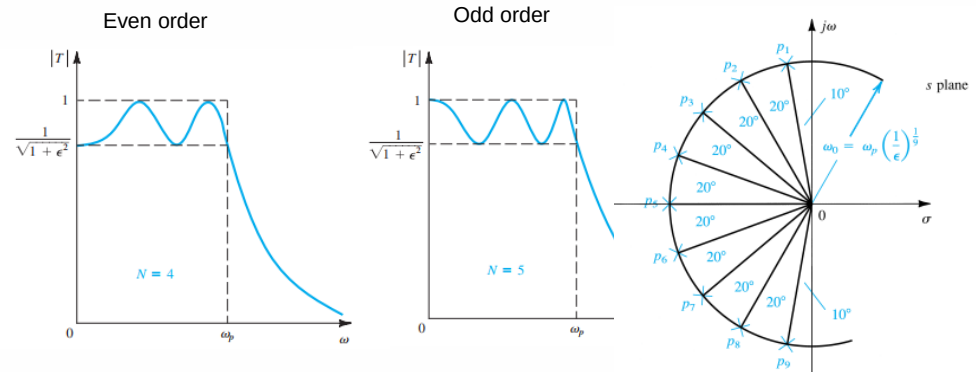


Chebyshev Filter

- The magnitude of the transfer function of an N th-order Chebyshev filter with a passband edge (ripple bandwidth) ω_p is given by:

- parameter ϵ gives the passband ripple:
- Attenuation at the stopband edge ($\omega = \omega_s$):

- Design procedure
 - Find ϵ from (1)
 - Find N from (2)
 - Determine the poles position from (3)
 - Write the transfer function (4)



$$|T(j\omega)| = \frac{1}{\sqrt{1 + \epsilon^2 \cos^2[N \cos^{-1}(\omega/\omega_p)]}} \quad \text{for } \omega \leq \omega_p$$

$$|T(j\omega)| = \frac{1}{\sqrt{1 + \epsilon^2 \cosh^2[N \cosh^{-1}(\omega/\omega_p)]}} \quad \text{for } \omega \geq \omega_p$$

$$(1) \quad A_{\max} = 10 \log(1 + \epsilon^2)$$

$$(2) \quad A(\omega_s) = 10 \log[1 + \epsilon^2 \cosh^2(N \cosh^{-1}(\omega_s/\omega_p))]$$

$$(3) \quad p_k = -\omega_p \sin\left(\frac{2k-1}{N} \frac{\pi}{2}\right) \sinh\left(\frac{1}{N} \sinh^{-1} \frac{1}{\epsilon}\right) + j\omega_p \cos\left(\frac{2k-1}{N} \frac{\pi}{2}\right) \cosh\left(\frac{1}{N} \sinh^{-1} \frac{1}{\epsilon}\right) \quad k = 1, 2, \dots, N$$

$$(4) \quad T(s) = \frac{K \omega_p^N}{\epsilon 2^{N-1} (s - p_1)(s - p_2) \cdots (s - p_N)}$$

Active 1'st order filters

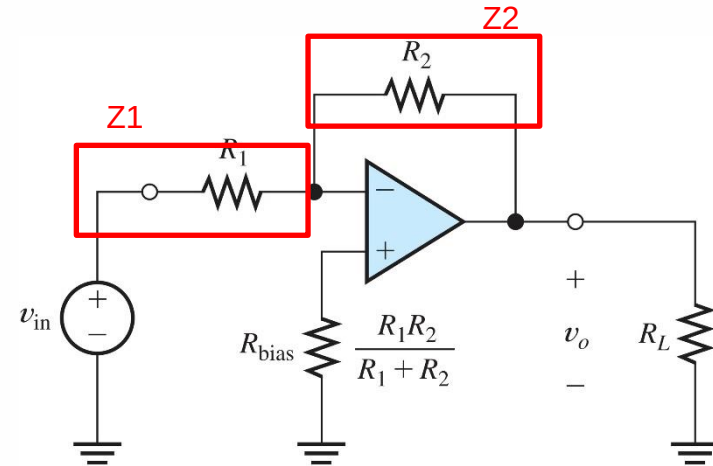
- Inverting amplifier:

- $$Av = -\frac{R_2}{R_1} = -\frac{Z_2}{Z_1}$$

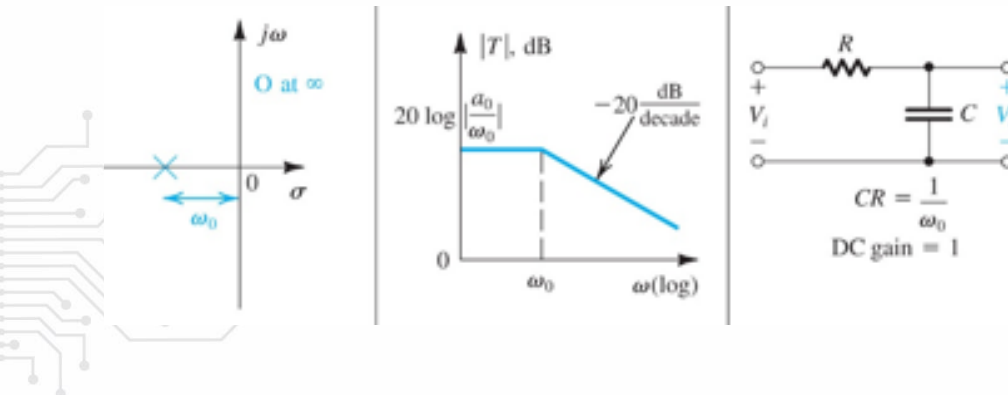
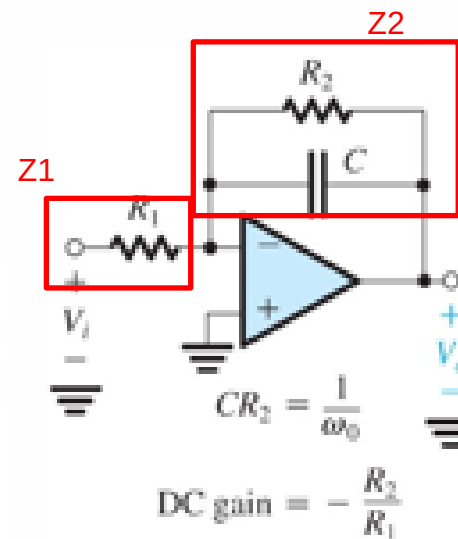
- Let's analyse this circuit:

- $$Z_2 = \frac{R_2 \frac{1}{Cs}}{R_2 + \frac{1}{Cs}} \text{ and } Z_1 = R_1$$

- $$Av = -\frac{Z_2}{Z_1} = -\frac{R_2}{R_1(R_2Cs+1)} = -\frac{R_2}{R_1} \frac{1}{(R_2Cs+1)}$$

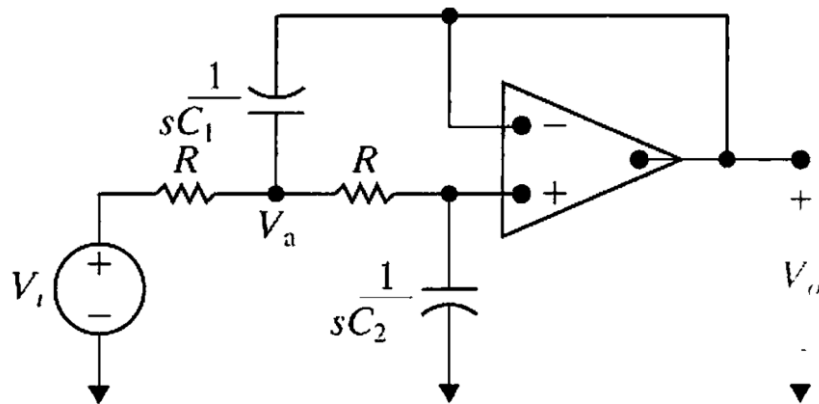


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Example: Butterworth Lowpass Filter

Design a fourth-order Butterworth low-pass filter with a cutoff frequency of 500 Hz and a passband gain of 1



$$\frac{V_a - V_i}{R} + (V_a - V_o)sC_1 + \frac{V_a - V_o}{R} = 0,$$

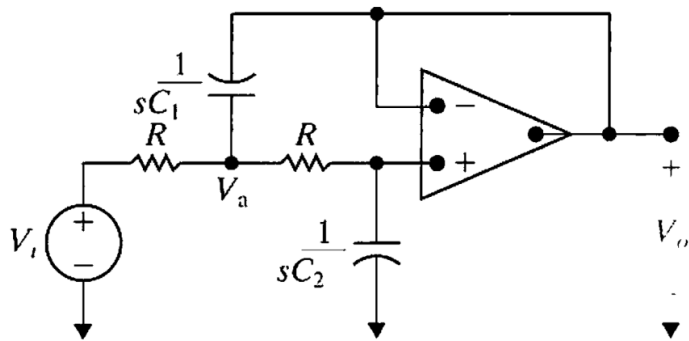
$$V_o sC_2 + \frac{V_o - V_a}{R} = 0.$$

$$H(s) = \frac{V_o}{V_i} = \frac{\frac{1}{R^2 C_1 C_2}}{s^2 + \frac{2}{RC_1}s + \frac{1}{R^2 C_1 C_2}}.$$

Finally, set $R = 1 \Omega$

$$H(s) = \frac{\frac{1}{C_1 C_2}}{s^2 + \frac{2}{C_1}s + \frac{1}{C_1 C_2}}.$$

Example: Butterworth Lowpass Filter



$$H(s) = \frac{\frac{1}{C_1 C_2}}{s^2 + \frac{2}{C_1} s + \frac{1}{C_1 C_2}}$$

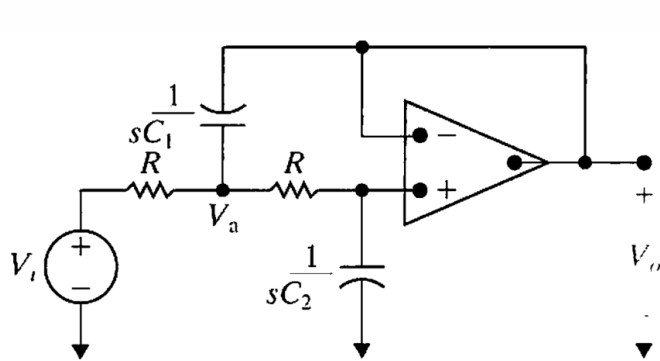
$$\frac{1}{s^2 + 0.765s + 1}$$

TABLE 15.1 Normalized (so that $\omega_c = 1$ rad/s) Butterworth Polynomials up to the Eighth Order

n	n th-Order Butterworth Polynomial
1	$(s + 1)$
2	$(s^2 + \sqrt{2}s + 1)$
3	$(s + 1)(s^2 + s + 1)$
4	$(s^2 + 0.765s + 1)(s^2 + 1.848s + 1)$
5	$(s + 1)(s^2 + 0.618s + 1)(s^2 + 1.618s + 1)$
6	$(s^2 + 0.518s + 1)(s^2 + \sqrt{2}s + 1)(s^2 + 1.932s + 1)$
7	$(s + 1)(s^2 + 0.445s + 1)(s^2 + 1.247s + 1)(s^2 + 1.802s + 1)$
8	$(s^2 + 0.390s + 1)(s^2 + 1.111s + 1)(s^2 + 1.6663s + 1)(s^2 + 1.962s + 1)$

$$(s^2 + 0.765s + 1)(s^2 + 1.848s + 1)$$

Example: Butterworth Lowpass Filter



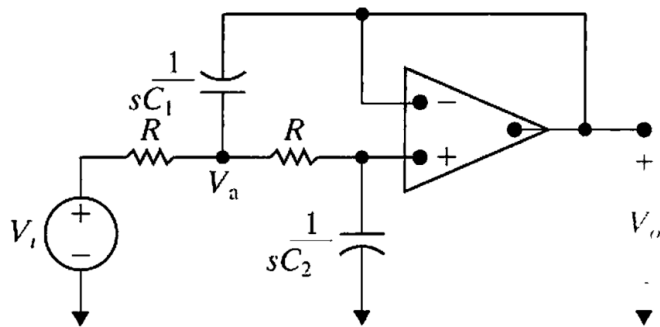
$$H(s) = \frac{\frac{1}{C_1 C_2}}{s^2 + \frac{2}{C_1} s + \frac{1}{C_1 C_2}}$$

$$\frac{K}{s^2 + 0.765s + 1}$$

$$\frac{2}{C_1} = 0.765 \Rightarrow C_1 = 2.6144$$

$$\frac{1}{C_1 C_2} = 1 \Rightarrow C_2 = 0.3825$$

Second stage:



$$\frac{1}{C_1 C_2} = 1 \Rightarrow C_2 = 0.3825$$

$$\frac{K}{s^2 + 1.848s + 1}$$

$$\frac{2}{C_1} = 1.848 \Rightarrow C_1 = 1.0823$$

$$\frac{1}{C_1 C_2} = 1 \Rightarrow C_2 = 0.9240$$

Example: Butterworth Lowpass Filter

- Now, we need to de-normalize the component values and the cut frequency.
- Frequency: $Kf = \frac{2\pi 500}{1}$
- We chose R to 1k Ω , thus:
- $Kr = \frac{1000}{1}$

$$C'_1 = \frac{C_1}{K_f K_r} = \frac{2.6144}{2\pi 500 * 1000} = 830 \text{ nF}$$

$$C'_2 = \frac{C_2}{K_f K_r} = \frac{0.3825}{2\pi 500 * 1000} = 122 \text{ nF}$$

Second stage:

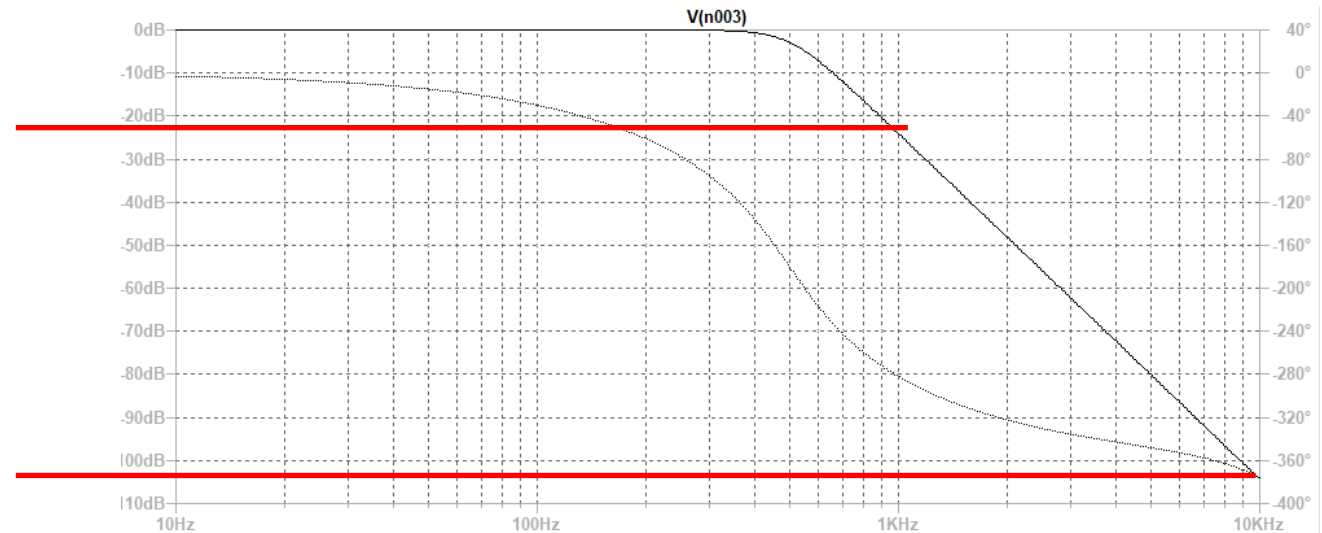
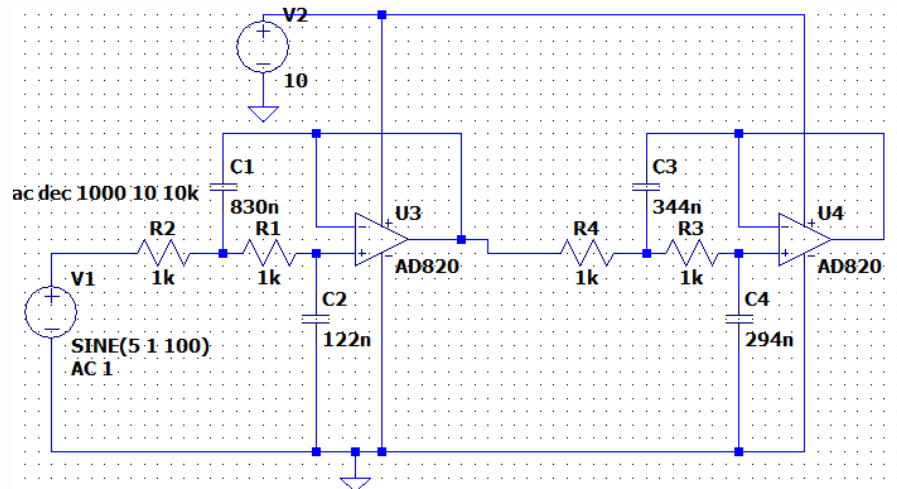
$$C'_1 = \frac{C_1}{K_f K_r} = \frac{1.0823}{2\pi 500 * 1000} = 344 \text{ nF}$$

$$C'_2 = \frac{C_2}{K_f K_r} = \frac{0.9240}{2\pi 500 * 1000} = 294 \text{ nF}$$

Let's simulate it in LT-spice...

Example: Butterworth Lowpass Filter

Design a fourth-order Butterworth low-pass filter with a cutoff frequency of 500 Hz and a passband gain of 1



Equal-component Sallen–Key lowpass active-filter section.

- Example 4'th order Butterworth with $f_c=100\text{Hz}$.

- Chose C and R_f , e.g.
 $C=100\text{nF}$ $R_f=10\text{k}\Omega$

- Find R:

$$R = \frac{1}{2\pi \cdot 100 \cdot 100\text{nF}} = 15.8\text{k}\Omega$$

- Find $(K-1)R_f$ - first section

$$(K-1)R_f = (1.152 - 1) \cdot 10\text{k}\Omega$$

- Find $(K-1)R_f$ - second section

$$(K-1)R_f = (2.235 - 1) \cdot 10\text{k}\Omega$$

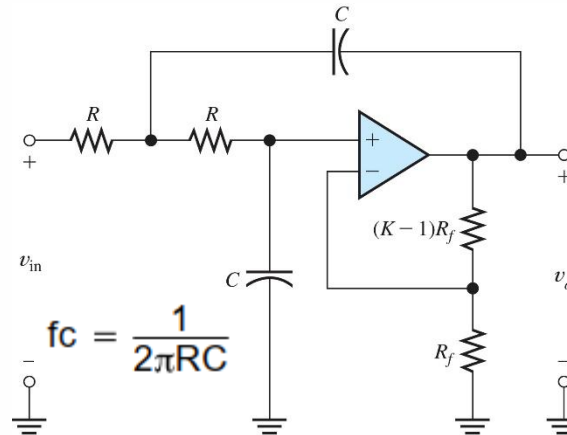
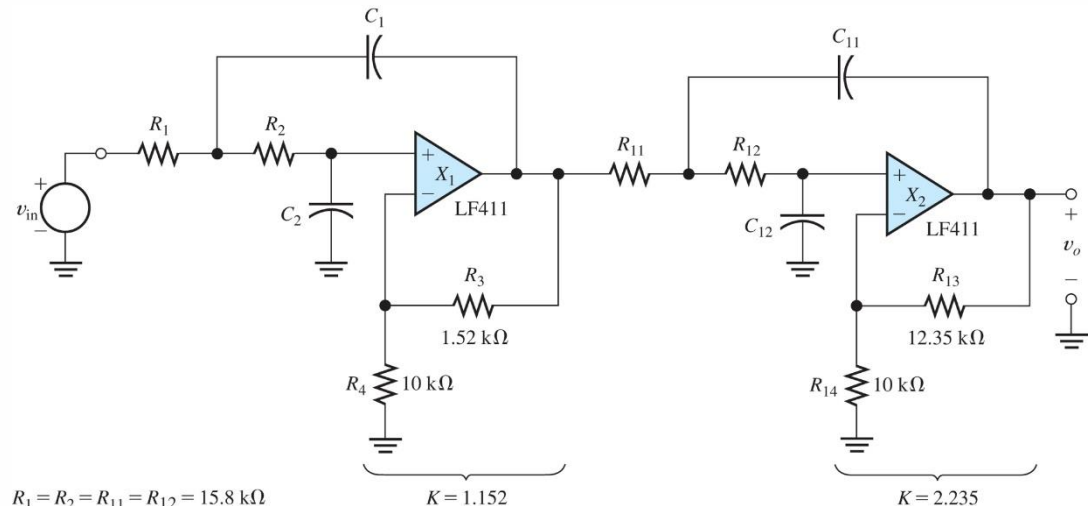


Table 13.1 *K* Values for Lowpass or Highpass Butterworth Filters of Various Orders

Order	<i>K</i>
2	1.586
4	1.152
6	2.235
8	1.068
	1.586
	2.483
	1.038
	1.337
	1.889
	2.610

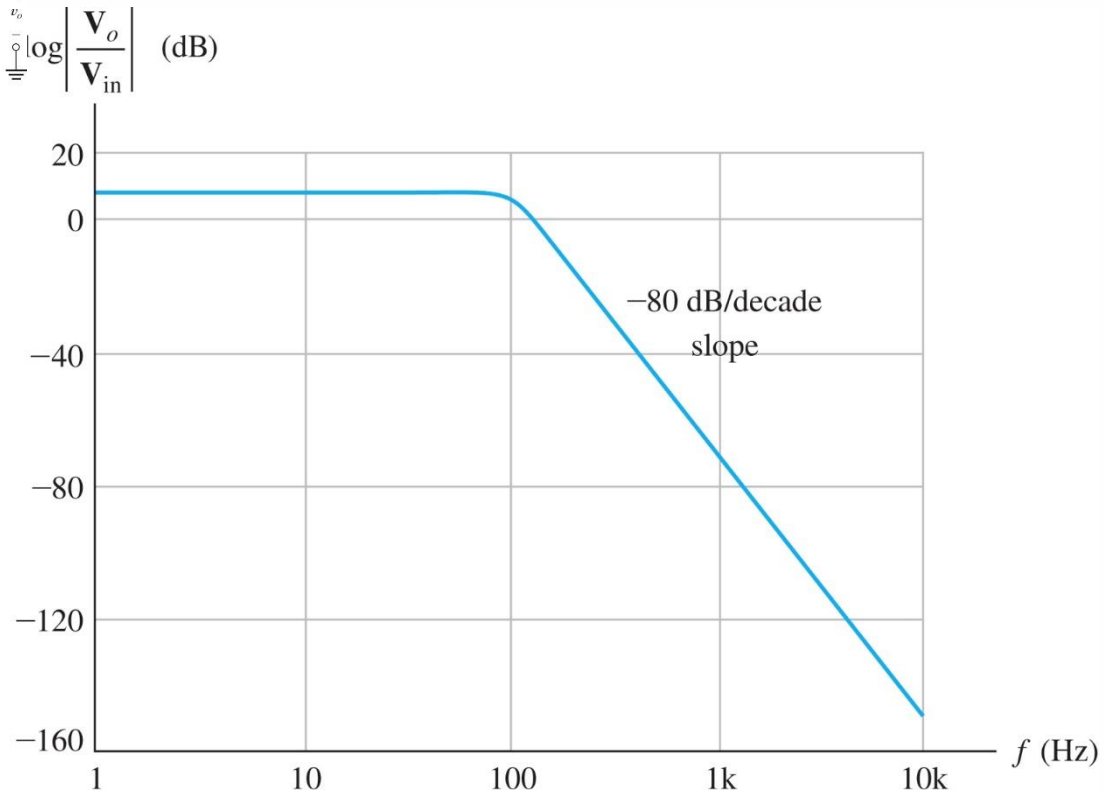
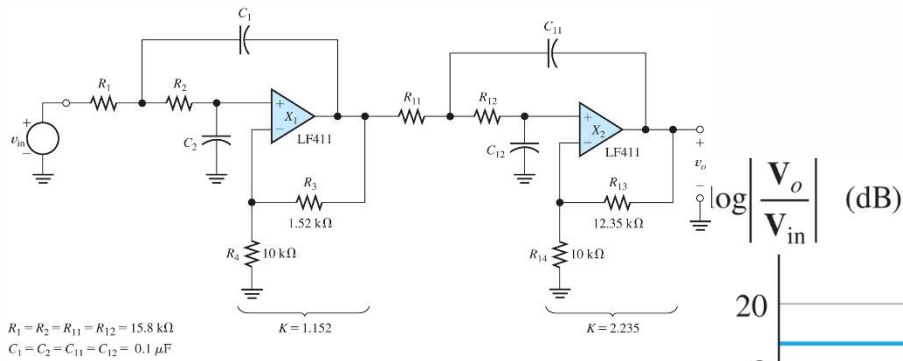


$$R_1 = R_2 = R_{11} = R_{12} = 15.8\text{ k}\Omega$$

$$C_1 = C_2 = C_{11} = C_{12} = 0.1\text{ }\mu\text{F}$$

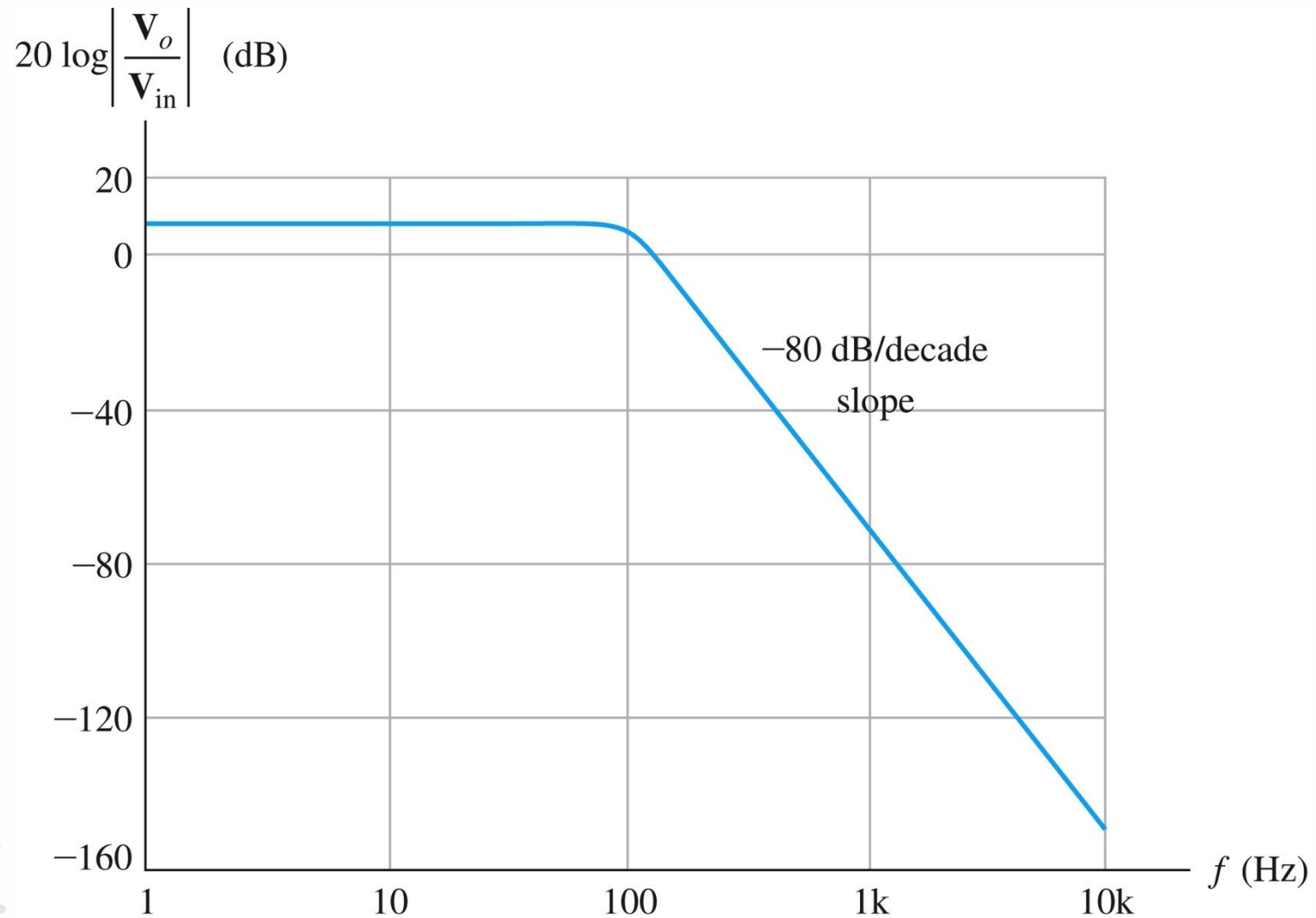
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Equal-component Sallen–Key lowpass active-filter section.



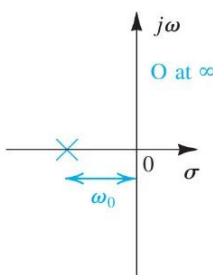
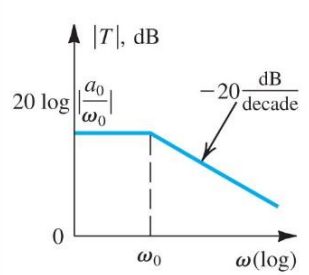
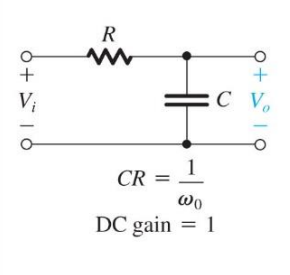
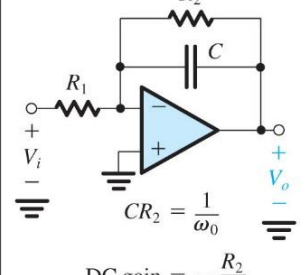
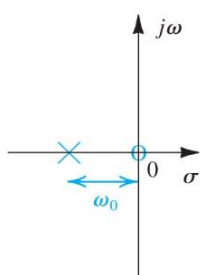
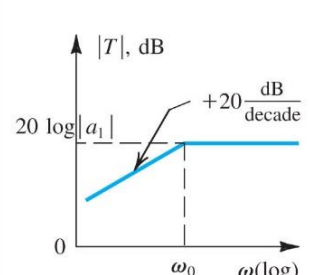
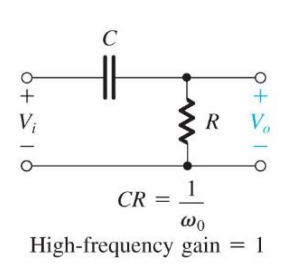
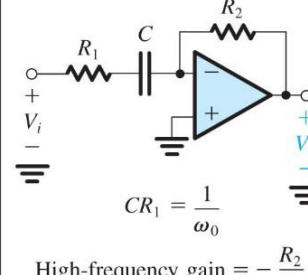
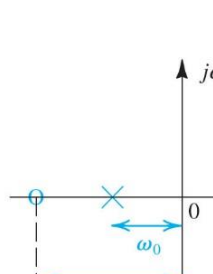
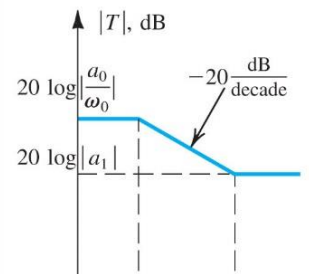
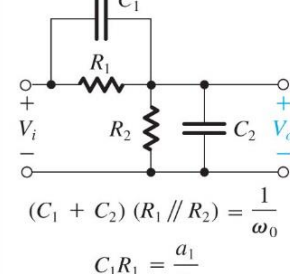
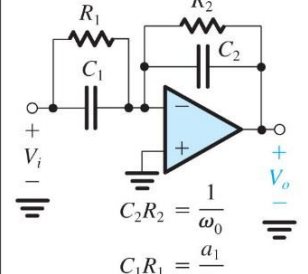
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Figure 13.42 Bode magnitude plot of the gain for the fourth-order lowpass filter of Example 13.8.



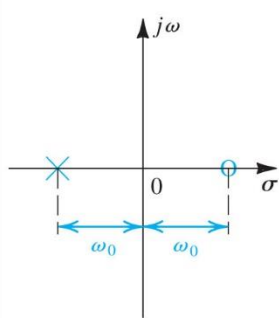
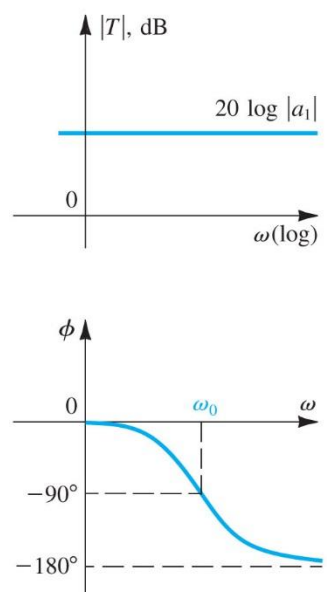
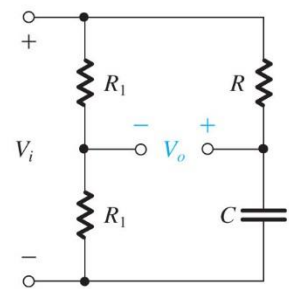
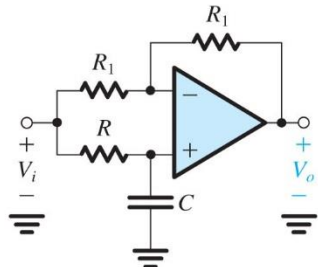
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Active filters

Filter Type and $T(s)$	s -Plane Singularities	Bode Plot for $ T $	Passive Realization	Op Amp-RC Realization
(a) Low pass (LP) $T(s) = \frac{a_0}{s + \omega_0}$			 $CR = \frac{1}{\omega_0}$ DC gain = 1	 $CR_2 = \frac{1}{\omega_0}$ DC gain = $-\frac{R_2}{R_1}$
(b) High pass (HP) $T(s) = \frac{a_1 s}{s + \omega_0}$			 $CR = \frac{1}{\omega_0}$ High-frequency gain = 1	 $CR_1 = \frac{1}{\omega_0}$ High-frequency gain = $-\frac{R_2}{R_1}$
(c) General $T(s) = \frac{a_1 s + a_0}{s + \omega_0}$			 $(C_1 + C_2)(R_1 \parallel R_2) = \frac{1}{\omega_0}$ $C_1 R_1 = \frac{a_1}{a_0}$ DC gain = $\frac{R_2}{R_1 + R_2}$ HF gain = $\frac{C_1}{C_1 + C_2}$	 $C_2 R_2 = \frac{1}{\omega_0}$ $C_1 R_1 = \frac{a_1}{a_0}$ DC gain = $-\frac{R_2}{R_1}$ HF gain = $-\frac{C_1}{C_2}$

Active filters



$T(s)$	Singularities	$ T $ and ϕ	Passive Realization	Op Amp-RC Realization
All pass (AP) $T(s) = -a_1 \frac{s - \omega_0}{s + \omega_0}$ $a_1 > 0$			 $CR = 1/\omega_0$ Flat gain $(a_1) = 0.5$	 $CR = 1/\omega_0$ Flat gain $(a_1) = 1$ $\left \frac{V_o}{V_i} \right = 1$ $\phi(\omega) = -2 \tan^{-1}(\omega CR)$

First order all-pass filter

